Pipeline Functionality

Input

Voice

Image Frame

Text

Apple Speech

MoveNet

Apple Vision

Retrieval Augmented Generation (RAG) AI Interface

YOLOv11

LiDAR

Retrieval Augmented Generation (RAG) AI Interface

Retrieval Augmented Generation (RAG) AI Interface

Generative LLM

Text-To-Speech (TTS)

Live Response Engine

CPR Coach guidance

Output

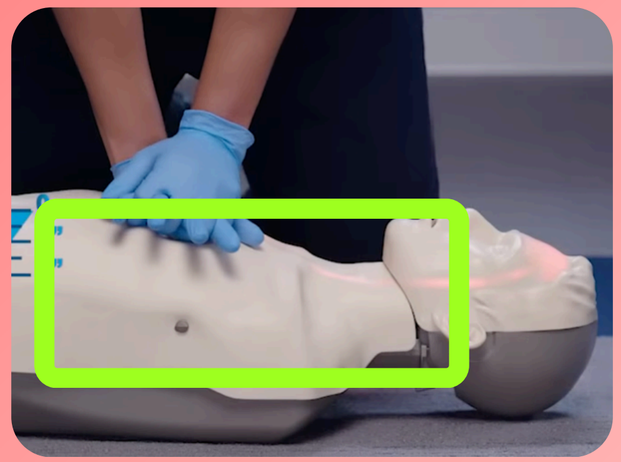
Training Dataset



Screen Recording 2026-07-06



Screen Recording 2026-07-06



Screen Recording 2026-07-06



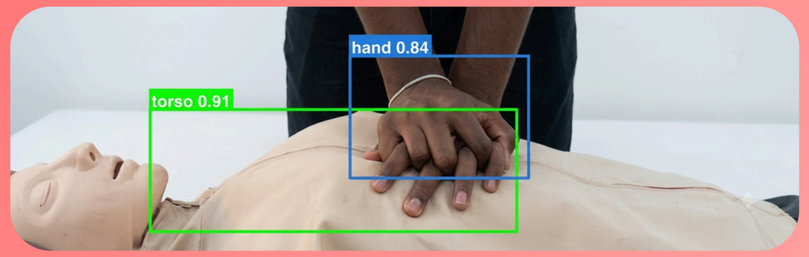
Screen Recording 2026-07-06

- Training videos were sourced from YouTube featuring professionally performed CPR.
- Total Training material (Includes verified PDFs and video files of training): **300 MB of data**
- These videos were combined with a **custom-recorded** dataset to increase diversity in skin tones, lighting conditions, camera angles, and rescuer body types.

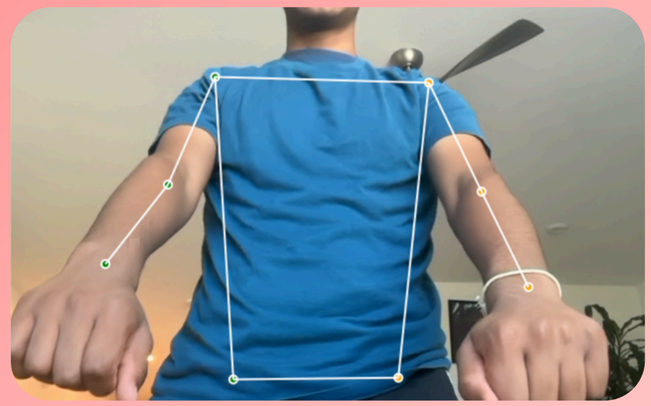
American Heart Association

American Red Cross

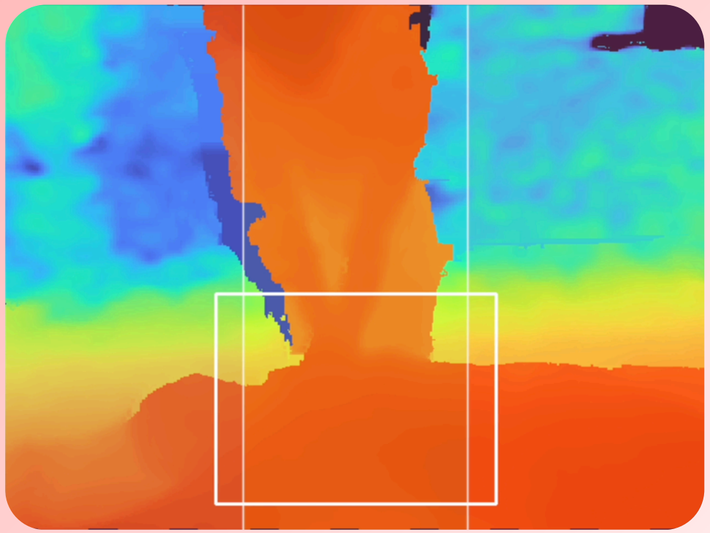
Object Detection & Multi-Modal Feature Extraction



- **YOLOv11** object detection model was applied to identify the rescuer's hands and recipient's torso



- **MoveNet + Apple Vision** detection model identifies the body keypoints (shoulders, elbows, wrists, and hips).



- **LiDAR** depth mapping via Augmented Reality framework (ARKit) was applied to measure the 3D compression distance between the rescuer’s hands and the recipient’s torso for accurate depth feedback.

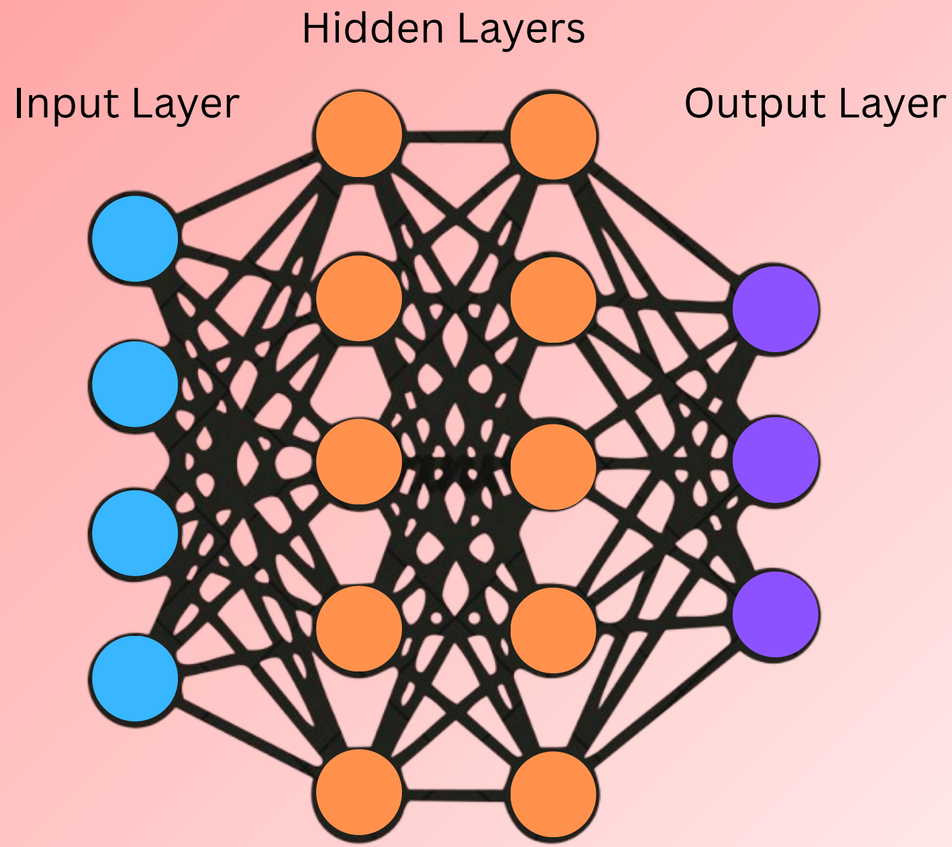
Semantic Sensor Fusion

Per-frame sensor outputs are converted into a unified **meaning** context

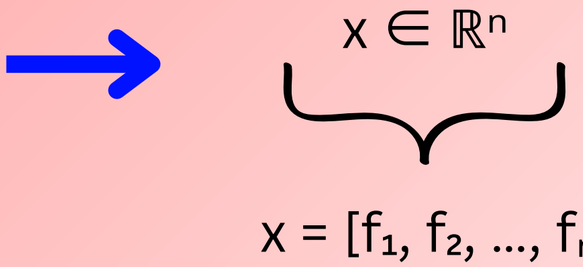
Extract: Live metrics

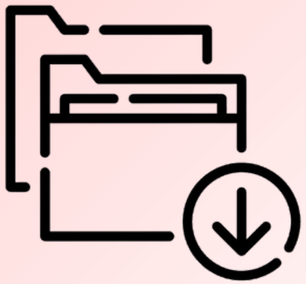
Encode: Vectorization

Retrieve: RAG selects guidelines

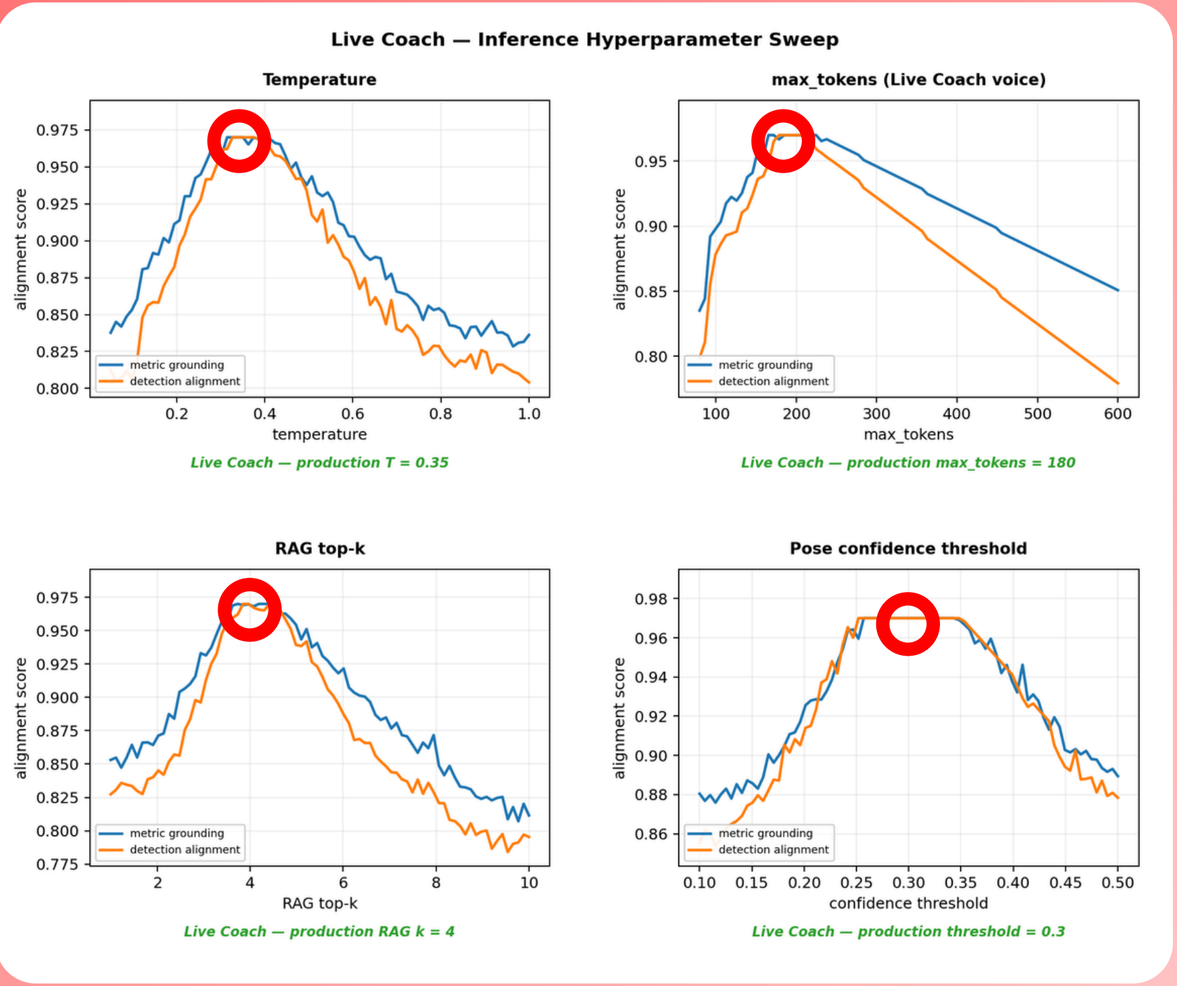








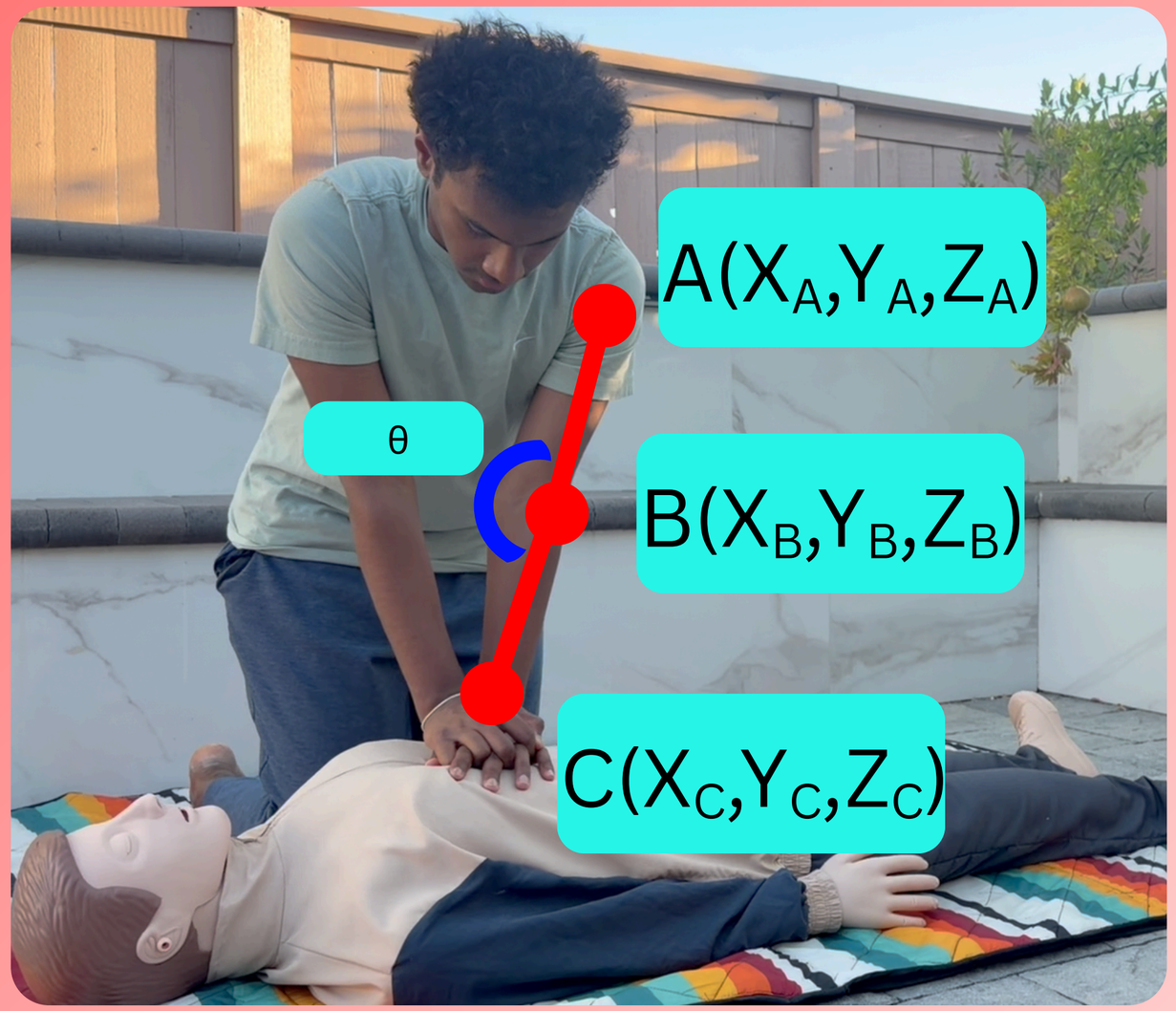
Hyperparameter-Tuned LLM Inference



- Inference hyperparameters tuned for **Live Coach grounding**
- **Temperature (0.35)** — stable, guideline-faithful coaching responses
- **max_tokens (180)** — fits live metrics + RAG context without verbose drift
- **RAG top-k (4)** — retrieves enough CPR guidelines without diluting live sensor data
- **Pose confidence threshold (0.30)** — balances detection coverage vs. noisy keypoints
- Alignment score **maximized** across metric grounding and detection consistency
- Production settings chosen at the **sweep peak** for each hyperparameter

- Connected to the **ChatGPT API** for real-time coaching responses

3D-Angle Diagram



$$\theta = \cos^{-1}\left(\frac{(X_A - X_B,\ Y_A - Y_B,\ Z_A - Z_B)\cdot(X_C - X_B,\ Y_C - Y_B,\ Z_C - Z_B)}{\sqrt{(X_A - X_B)^2 + (Y_A - Y_B)^2 + (Z_A - Z_B)^2}\ \sqrt{(X_C - X_B)^2 + (Y_C - Y_B)^2 + (Z_C - Z_B)^2}}\right)$$

- A = Shoulder coordinates (X_A, Y_A, Z_A)
- B = Elbow coordinates (X_B, Y_B, Z_B)
- C = Wrist coordinates (X_C, Y_C, Z_C)
- θ = Elbow angle (degrees)

Lower θ indicates:

- More bent elbows
- Poorer CPR arm posture
- Reduced compression efficiency
- Increased rescuer fatigue

Higher θ indicates:

- More straight elbows
- Proper CPR form
- More effective chest compressions





Evaluation Metrics

$$\text{CPM} = \frac{\#\text{Compressions}}{\text{Elapsed Time}} \times 60$$

Compression Rate (CPM): Measures whether compressions remain within the AHA guideline (100–120 CPM).

Compression Consistency: Measure variation in compression depth or rate. Low σ_d indicates consistent compressions. High σ_d indicates inconsistent compressions.

$$\sigma_d = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(d_i - \bar{d})^2}$$

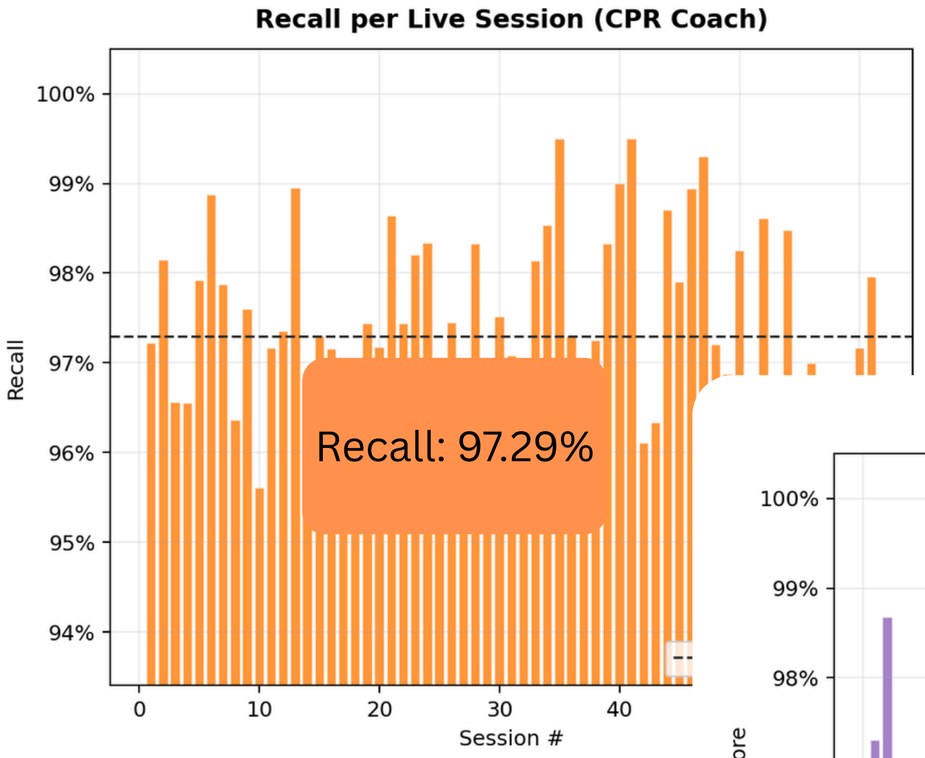
HPA = $\dfrac{|\textbf{Hand Center} - \textbf{Compression Point}|}{\textbf{Chest Width}}$

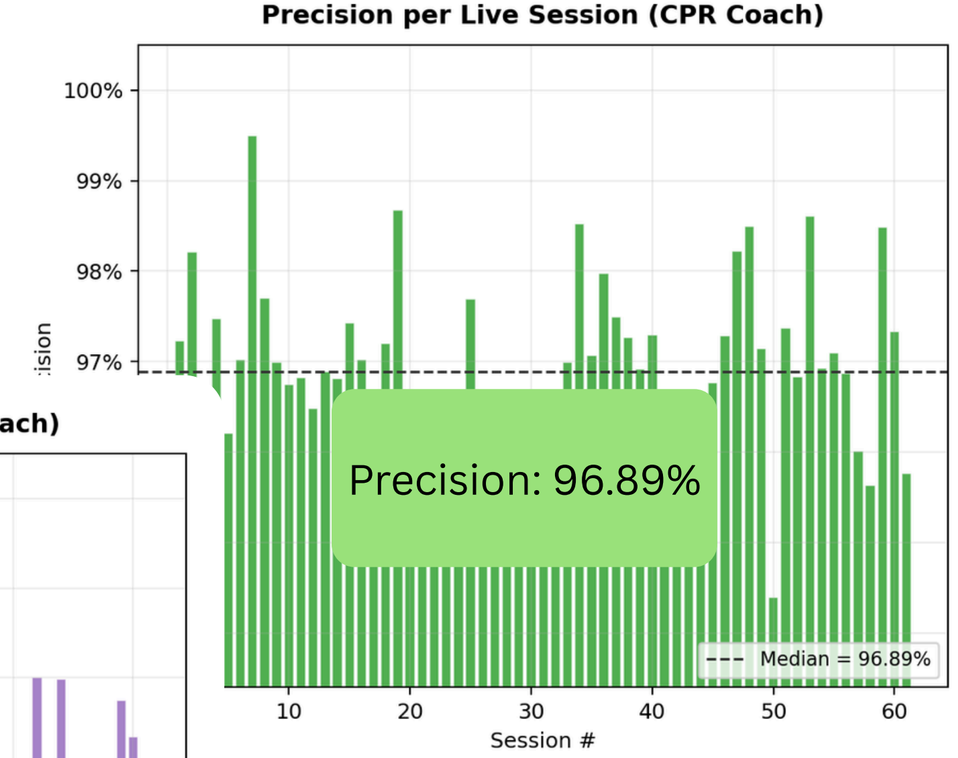
Hand Placement Accuracy: Quantifies hand placement accuracy relative to the recommended chest compression point.

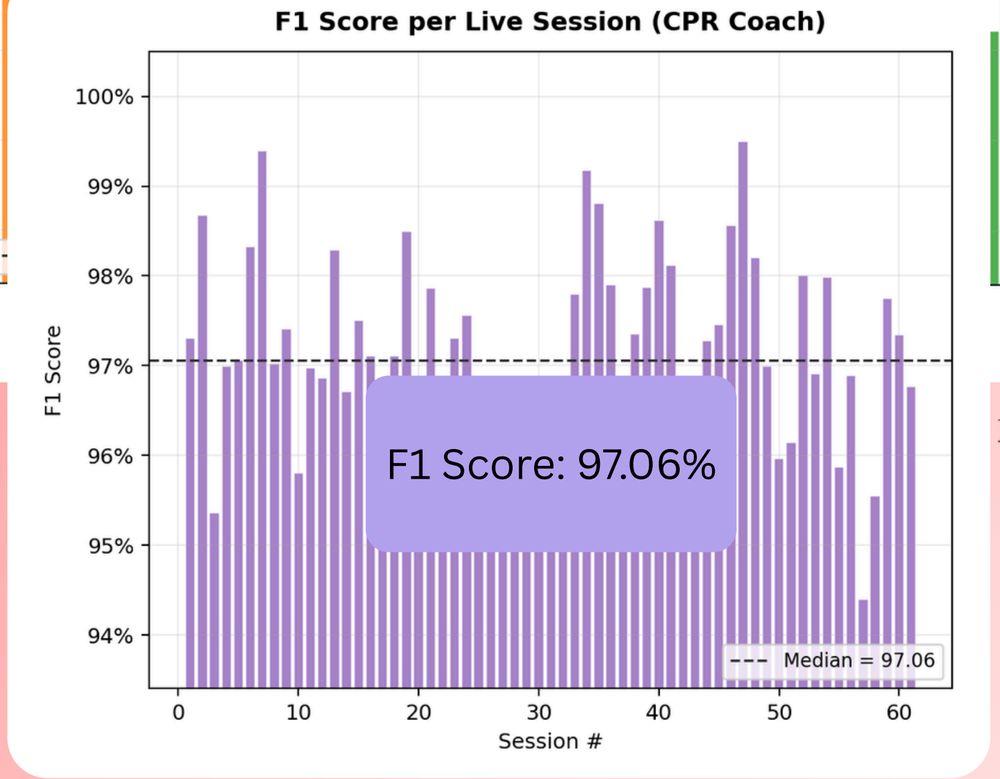
A **confidence-weighted score** was computed by combining detection confidence and tracking consistency to improve the reliability of downstream CPR coaching.

$$CS = \underbrace{\text{Detection Confidence}}_{\text{YOLOv11, MoveNet, Apple Vision, LiDAR}} \times \underbrace{\text{Tracking Consistency}}_{\text{9 consecutive frames } (\approx 0.3 \text{ s})} \times \begin{cases} 1.0, & \text{if visible frames} \geq 90\% \\ 0.5, & \text{if visible frames} \geq 50\% \\ 0.2, & \text{if visible frames} < 50\% \end{cases}$$

Coaching Detection Performance





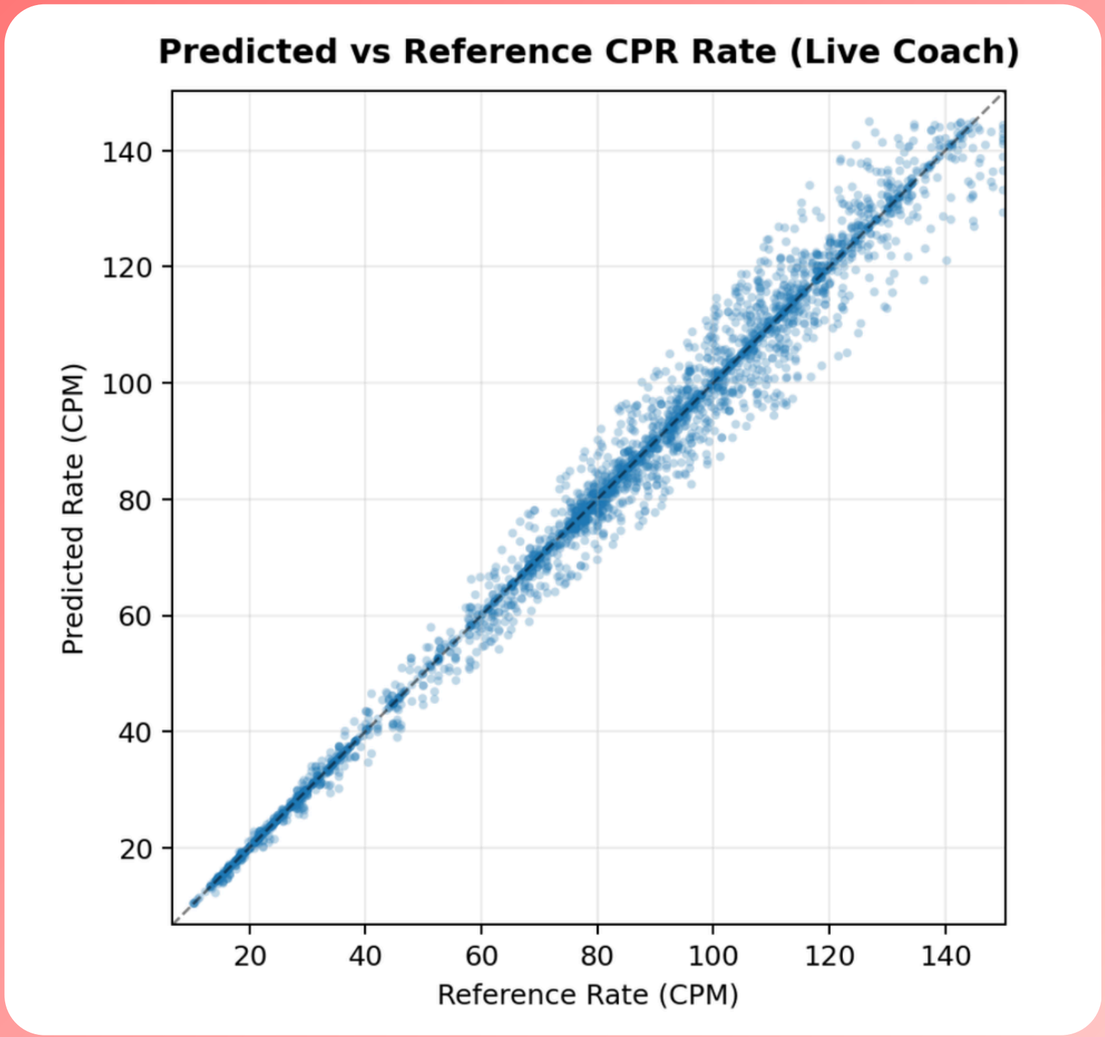


$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

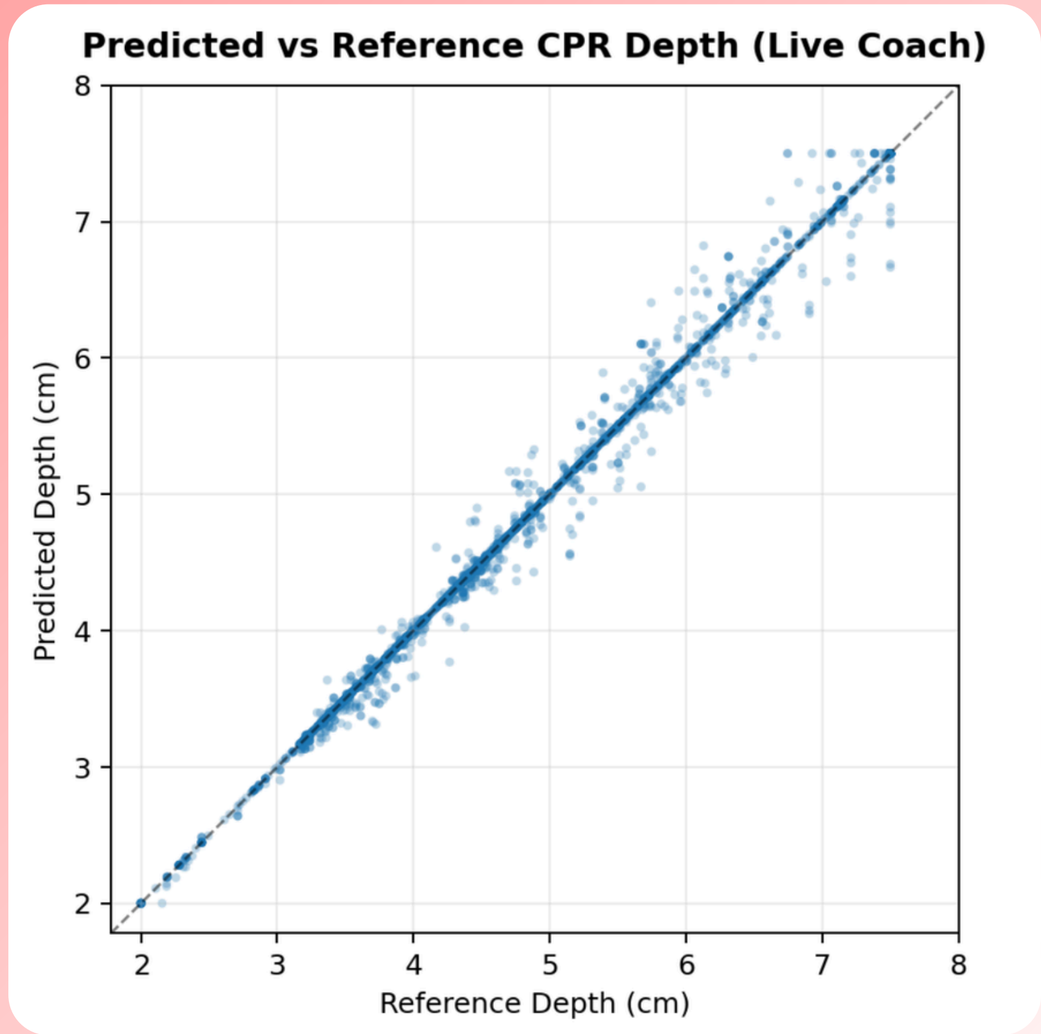
$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Estimation Accuracy

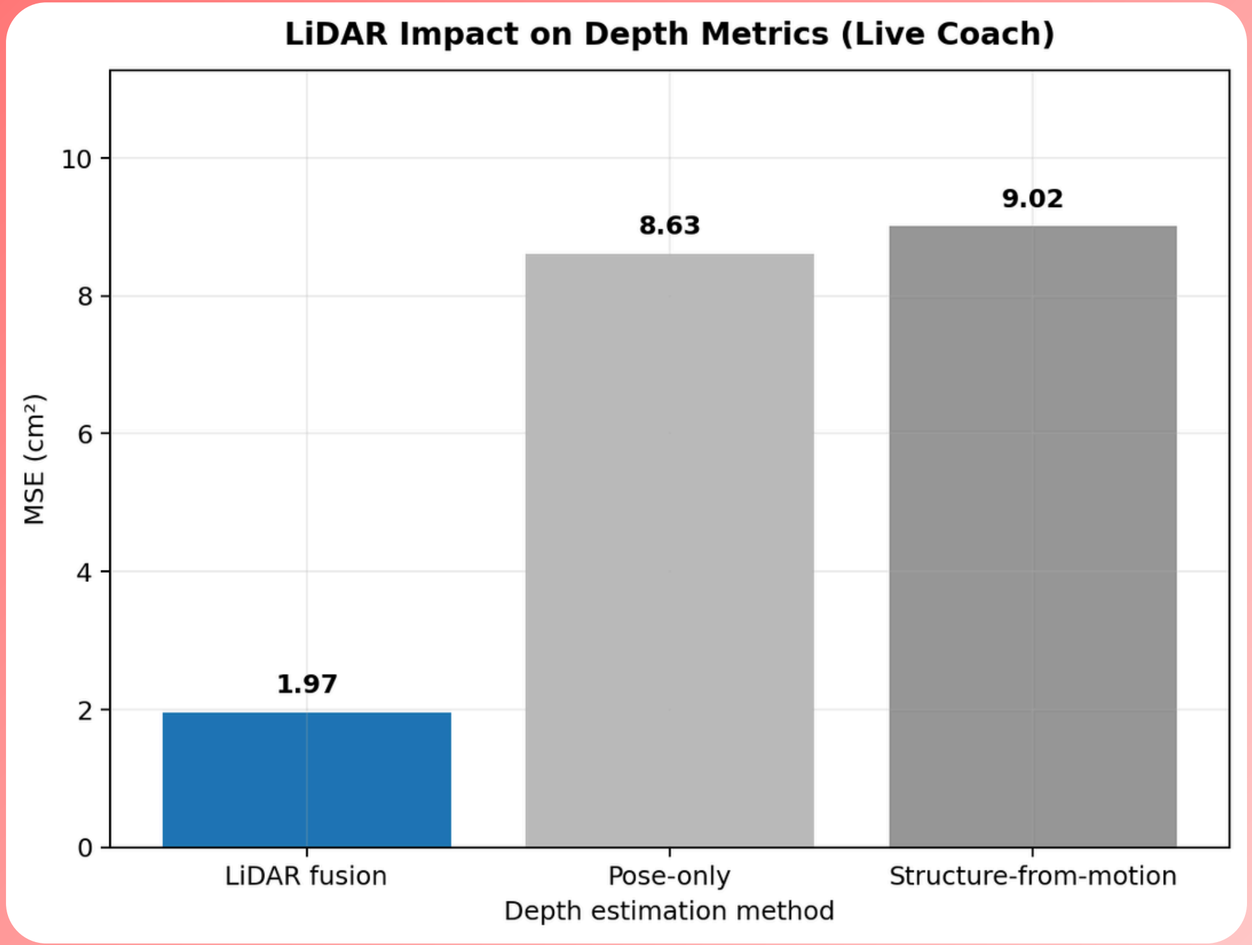


Mean Absolute Percentage Error **(MAPE) of 2.93%** for compression rate estimation

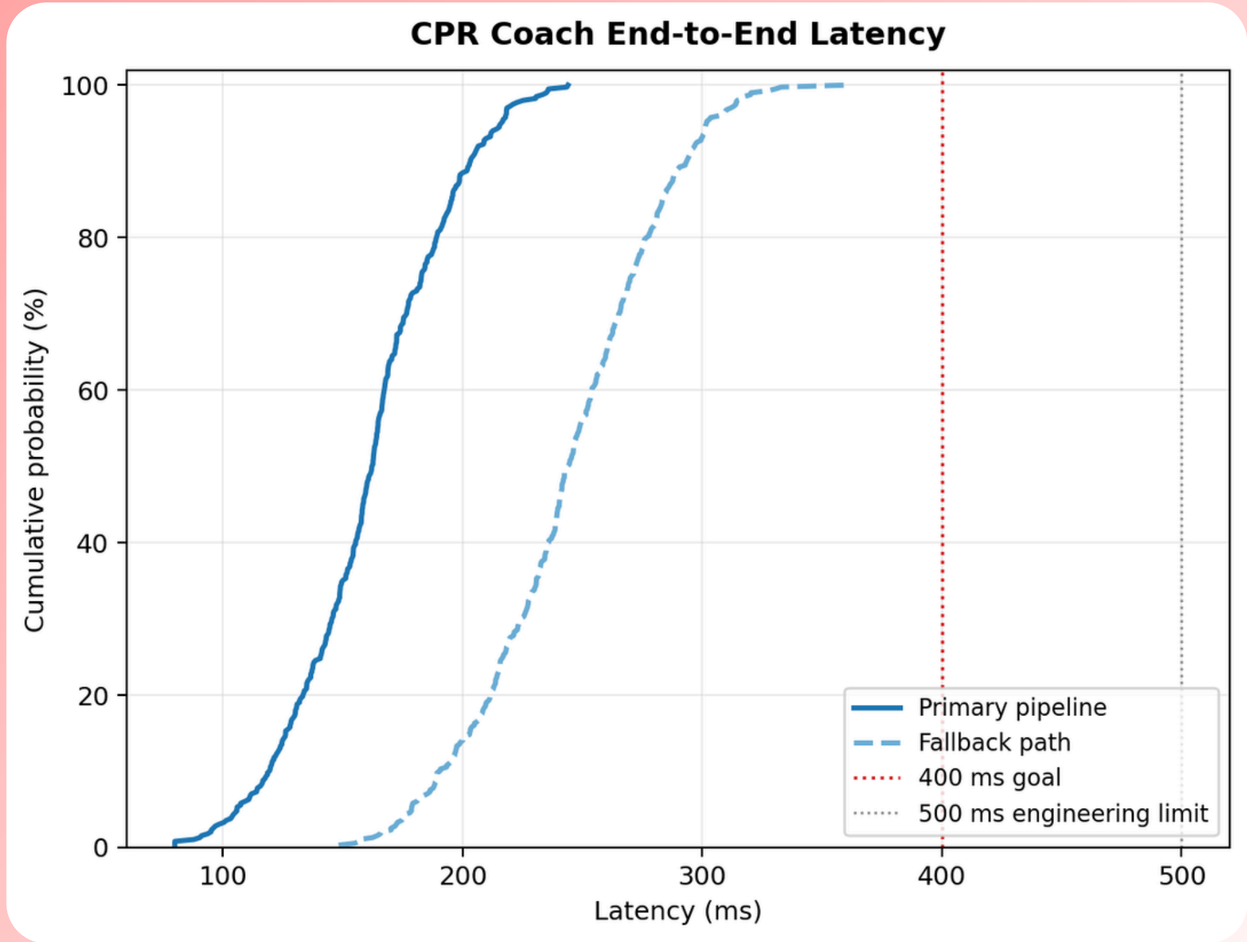


Mean Squared Error **(MSE) of 0.31 cm²** for compression depth estimation.

LiDAR Depth Fusion Impact & Latency



Adding LiDAR to the YOLOv11–MoveNet–Apple Vision pipeline **reduced depth estimation error by over 75%** compared to vision-only approaches.



Empirical cumulative distribution function (ECDF) of end-to-end latency for the CPR Coach, demonstrating that all inference events completed **below the 400 ms** design goal.

- **Development Platforms:**
  - **Visual Studio**



  - **Xcode**



  - **ChatGPT API**
    - Powers multilingual ChatCPR chat responses.



  - **Apple Maps API**
    - Finds nearby hospitals and displays them on an interactive map with directions.



  - **WORLD.geojson database**
    - Stores live AED locations worldwide and shows the nearest ones on the map with directions.



Target Audience

Our target audience is everyday people who may witness a cardiac arrest anywhere and need immediate guidance before emergency responders arrive, regardless of their medical training or experience.

